
Project NEXUS

Enterprise LLM Token Optimization, Governance & Resilient
Routing Engine

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Executive Summary & Core Objectives

STRATEGIC OVERVIEW

⚠️ The Enterprise Challenge

- **Runaway token consumption** from defaulting prompts to expensive flagship models.
- **Lack of accountability** and unallocated cloud spend without strict cost-center tracking.
- **Fragile dependencies** on upstream APIs prone to timeout and failure bottlenecks.

🛡️ The NEXUS Solution

- **Modular Architecture:** Decouples prompt classification and WBS validation from transport.
- **Mandatory Perimeter Defense:** Rejects unbilled requests before gateway contact.
- **Automated Resilience:** Implements exponential backoffs, circuit breakers, and local failovers.

Dynamic Multi-Tier Model Routing

[SRC/ORCHESTRATOR/ROUTER.PY](#)

Tier 1 `gpt-4o-mini`

Cost-optimized and high-throughput. Targets syntax checks, JSON parsing, chunking, and boilerplate extraction.

Tier 2 `gpt-4o`

Standard processing and reasoning. Handles complex debugging, architecture design, and system logic.

Tier 3 `o1-preview`

Frontier reasoning intelligence. Reserved strictly for statutory audits and complex legal/compliance analysis.

```
def classify_task(
    self,
    task_type: Optional[str],
    prompt: str = ""
) -> ModelTierConfig:
    # Check extreme complexity patterns first
    if any(re.search(p, task) for p in TIER_3_PATTERNS):
        return MODEL_TIERS["tier_3"]

    # Check high throughput patterns
    if any(re.search(p, task) for p in TIER_1_PATTERNS):
        return MODEL_TIERS["tier_1"]

    # Fallback to standard reasoning
    return MODEL_TIERS["tier_2"]
```

Governance & WBS Billing Enforcement

PERIMETER DEFENSE

Header Standards & Validation

- Accepts standardized corporate header aliases (X-EY-WBS-Element, X-Cost-Center, etc).
- Validates strict pattern matching using regex format: WBS-[A-Z0-9_-]+.

Perimeter Defense & APIM

- Requests lacking a valid WBS identifier are intercepted immediately.
- Rejected at the API boundary with an HTTP 400 error prior to touching models.

```
def prepare_request(self, request: object) -> PreparedRequest:
    # Extract corporate billing headers
    wbs = next((
        headers[name].strip() for name in (
            "X-EY-WBS-Element",
            "X-Cost-Center"
        ) if headers.get(name, "").strip()
    ), "")

    if not wbs:
        raise ValueError(
            "Missing mandatory X-EY-WBS-Element"
        )

    # Enforce strict WBS Regex contract
    if not re.fullmatch(r"WBS-[A-Z0-9_-]+", wbs):
        raise ValueError("Invalid WBS format")
```

Resilience, Retries & Local Fallback

SRC/ORCHESTRATOR/GATEWAY_CLIENT.PY

Retry Policy & Backoff

- Up to 3 automatic retry attempts on upstream connection or timeout failures.
- Exponential backoff timing (1s, 2s, 4s delay capping at 5s) to absorb temporary jitter.

Circuit Breaker & Fallback

- Automatically trips after consecutive upstream failures.
- Seamlessly reroutes traffic to the local sandbox (localhost:8080) for zero-downtime simulation.

```
try:
    payload = self._send_to(
        primary_url=self.primary_url,
        body=request_body,
        headers=prepared.headers,
    )
    self.circuit_breaker.record_success()
    return GatewayResponse(
        payload["model"], "primary", payload, telemetry
    )
except (HTTPError, URLError, TimeoutError, OSError):
    # Trip breaker & fallback to sandbox
    self.circuit_breaker.record_failure()
    return self._send_to_fallback(
        request_body, prepared.headers, telemetry
    )
```

Strict Telemetry & Verification Status

SRC/SCHEMAS/TELEMETRY.PY

Zero-Tolerance Telemetry

- Pydantic v2 contract strictly enforcing exactly 4 fields.
- Extra properties explicitly forbidden (extra="forbid") for audit-ready compliance.

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Automated Tests Passed

pytest -v in 0.08s

```
class TelemetryEvent(BaseModel):  
    """Audit event containing only required fields."""  
  
    # Forbid extra properties for compliance  
    model_config = ConfigDict(  
        extra="forbid",  
        strict=True,  
    )  
  
    timestamp: datetime  
  
    # Billing field must match exact Regex  
    wbsElement: str = Field(pattern=r"^WBS-[A-Z0-9_-]+$")  
  
    promptComplexity: Literal["low", "high"]  
  
    tokenCount: int = Field(ge=1)
```

Quality & Validation

Test Suite • 31 Tests • 99% Coverage

Five Coverage Areas

TEST MATRIX

Area	Tests	What's Covered
Routing Classification	7	Tier 1/2/3 dispatch, prompt semantics, priority override, unknown tasks.
WBS Perimeter Defense	3	Missing header rejected, invalid format rejected, valid WBS enriches headers.
Telemetry Schema	3	Required fields accepted, unknown fields forbidden, zero tokenCount rejected.
Gateway Resilience	5	Primary success, payload preservation, fallback, retry exhaustion.
Circuit Breaker	8	Threshold trip, auto-recovery, fast-fail, confidence SLA, state reset.

i All gateway I/O is mocked — no live network calls in the test suite.

31 Passed. 0 Failed.

PYTEST EXECUTION RESULTS

99%

TOTAL CODE COVERAGE

- **100%** Router Logic Tested
- **100%** Circuit Breaker SLA
- **100%** Telemetry Contracts

```
===== test session starts =====
platform darwin -- Python 3.11.0, pytest-7.4.3
rootdir: /enterprise/nexus/backend
collected 31 items

tests/test_router.py ..... [ 22%]
tests/test_apim_routing.py ... [ 32%]
tests/test_schemas.py ... [ 41%]
tests/test_resilience.py ..... [ 58%]
tests/test_circuit_breaker.py ..... [ 83%]
tests/test_nexus_runtime.py ..... [100%]

----- coverage: platform darwin, python 3.11.0-final-0 -----
Name                               Stmts  Miss  Cover
-----
src/orchestrator/router.py          42      0  100%
src/orchestrator/gateway_client.py  65      1   98%
src/orchestrator/circuit_breaker.py 38      0  100%
src/schemas/telemetry.py           18      0  100%
-----
TOTAL                               163      1   99%

===== 31 passed in 0.12s =====
```

Three Guarantees, Proven by Code

ENGINEERING ACCOUNTABILITY

01

Routing Tested

All three model tiers, classification paths, edge cases, and defaults verified.

02

Untagged Blocked

WBS enforcement strictly validated before any network socket is opened.

03

Failures Exited

Fallback scenarios, retry exhaustion, and circuit breaker recoveries exercised.

99% coverage isn't a vanity metric.

The one untested line is a logging branch, not a billing or routing path.